Energy Anomaly Detection System

This project introduces a Machine Learning-based Energy Anomaly Detection System designed to address the significant financial losses incurred by commercial buildings due to unexpected energy consumption spikes. Commercial buildings account for nearly 30% of global energy consumption, leading to over $200 billion annually in energy costs. Our system aims to detect abnormal usage, identify patterns, estimate financial impact, and generate actionable business recommendations through an interactive Streamlit Dashboard and automated PDF reports.

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Executive Summary

Cost Savings

Detecting anomalies prevents massive financial losses from equipment failure and inefficiencies.

Pattern Recognition

Identifies seasonal and temporal energy usage patterns for optimized operations.

Anomaly Detection

Machine Learning-based system detects abnormal energy usage with high accuracy.

Automated Reporting

Generates executive reports and business recommendations automatically.

The system processes over 800,000 time-series records, engineers more than 50 features, and implements multiple anomaly detection algorithms. It automatically switches models for large datasets and quantifies financial impact, achieving an approximate 5% anomaly detection rate, aligning with industry standards.

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Problem Statement & Objectives

The Challenge

Commercial energy systems generate vast amounts of time-series data from various sources like electricity, chilled water, gas, and weather parameters. Manually identifying anomalies in these high-volume datasets is time-consuming, error-prone, inefficient, and costly. This leads to significant financial losses due to equipment malfunction, system inefficiencies, and abnormal usage patterns.

Our Solution

We need an automated AI system that can effectively detect anomalies, classify abnormal patterns, quantify their cost impact, and support informed business decisions. Our system aims to process large datasets, engineer meaningful features, implement robust anomaly detection algorithms, and provide comprehensive reporting.

Technical Objectives

- Build robust preprocessing pipelines.
- Engineer 35+ meaningful time-series features.
- Train multiple anomaly detection models.
- Implement model switching for large datasets.
- Generate anomaly scores and provide visualization.

Business Objectives

- Quantify energy loss in monetary terms.
- Identify peak anomaly hours and analyze seasonal trends.
- Generate actionable recommendations for ROI-driven decisions.

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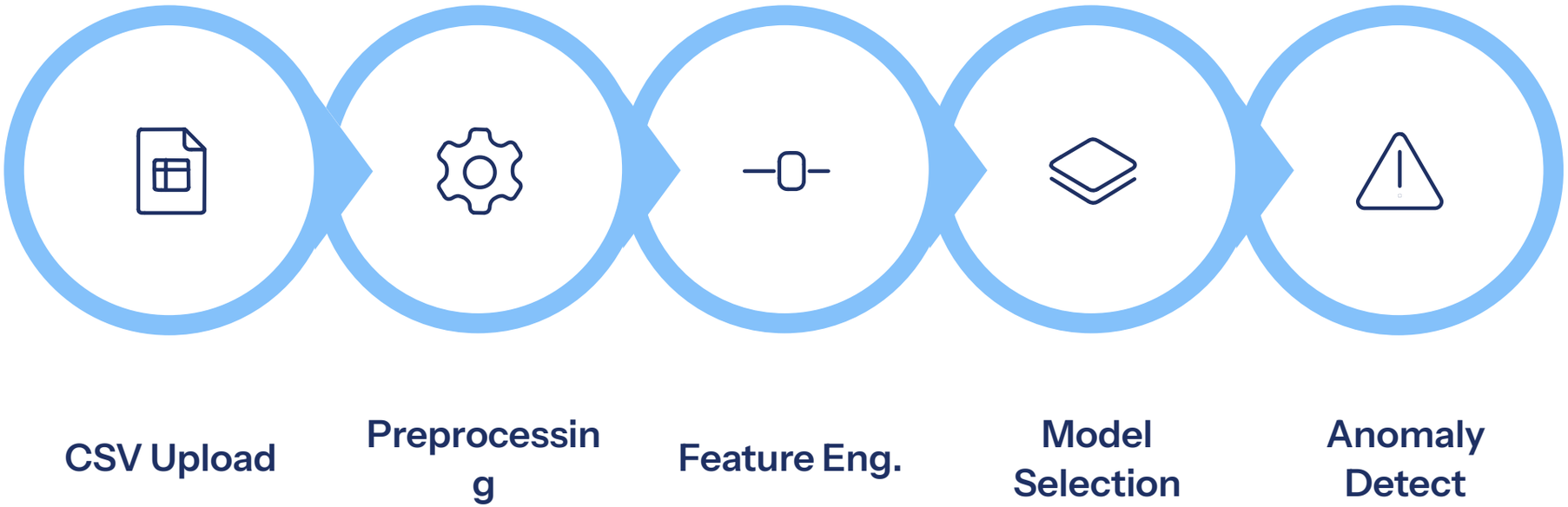
System Architecture & Data Overview



The system begins with CSV upload, followed by rigorous preprocessing and extensive feature engineering, creating over 50 features. Users can select models, or the system can auto-switch for large datasets, leading to anomaly detection, evaluation, and visualization. The final stages involve generating business insights and a comprehensive PDF executive report.

Dataset Overview

Our typical dataset comprises over 800,000 records with 9-12 raw features, consisting of hourly timestamped energy and weather variables. Key columns include electricity, chilled_water, gas, hot_water, water, temperature, humidity, and timestamp, providing a rich foundation for anomaly detection.

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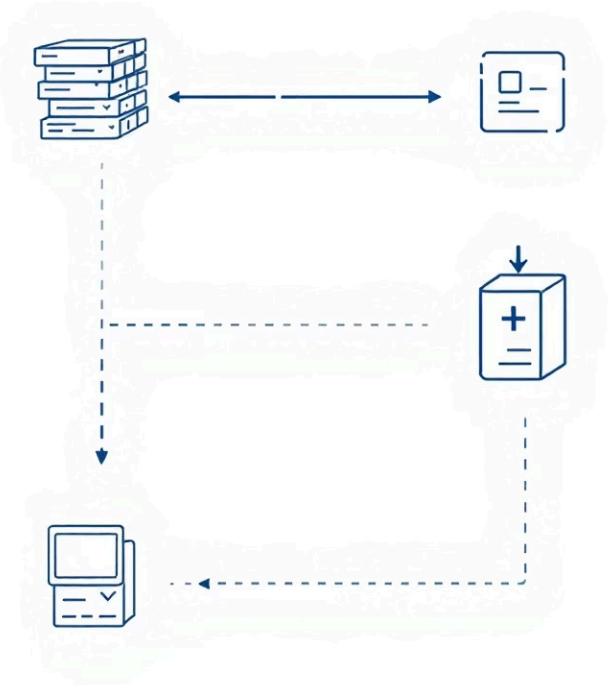
Data Preparation & Feature Engineering

Data Preprocessing

Our preprocessing pipeline ensures data quality and readiness for machine learning. This involves converting timestamps to datetime objects, sorting data by time, and handling missing values through forward and backward filling. We cap outliers at the 99th percentile to prevent extreme spikes from skewing models and scale all numerical features to a [0, 1] range using MinMaxScaler. The system also automatically detects energy columns.



Feature Engineering (50+ Features)

We engineer a comprehensive set of features to capture various aspects of energy consumption patterns:

- **Temporal Features:** Hour, Day, Month, Quarter, Week, Day of Week, Weekend Flag.
- **Cyclical Encoding:** hour_sin/cos, month_sin/cos to capture cyclical patterns.
- **Rolling Statistics (5 Windows):** Rolling Mean, Std, Min, Max over 6, 12, 24, 48, 168 hours.
- **Lag Features (10 Lags):** lag_1, lag_2, lag_3, lag_6, lag_12, lag_24, lag_48, lag_72, lag_168, lag_336 to capture past dependencies.
- **Deviation Metrics:** zscore_24, zscore_168 to quantify abnormal deviations from baseline.

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Machine Learning Models & Output

Isolation Forest

Ideal for large datasets and high-dimensional data, offering efficient anomaly detection.

Local Outlier Factor (LOF)

A density-based approach, effective for identifying local anomalies.

One-Class SVM

A boundary-based model that learns the boundary of normal data points.

Robust Covariance

Utilizes an elliptic envelope to detect anomalies based on Gaussian distribution assumptions.

The system incorporates smart model switching: if a dataset exceeds 200,000 rows, it automatically defaults to Isolation Forest due to its superior scalability. Each sample processed by the models receives a final anomaly classification (0 or 1) and an anomaly score, with a typical anomaly rate of approximately 5% across 800,000 samples, resulting in around 40,000 detected anomalies.

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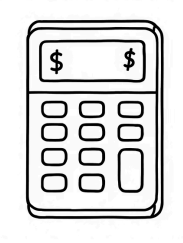
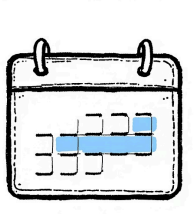
Business Insights & Impact

Our system translates complex ML results into tangible business value, focusing on financial impact and actionable recommendations.



Quantifying Cost Loss

The system calculates estimated financial loss by multiplying total anomalous energy consumption by the cost per unit. For instance, 300,000 units of anomalous consumption at $0.12 per unit translates to an estimated loss of $36,000.

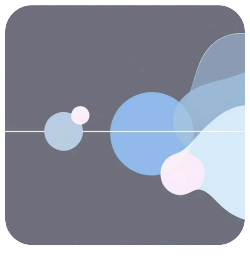


Seasonal Analysis

Identifies peak anomaly months and high-risk seasonal periods, helping to uncover weather-related abnormal patterns and inform preventive measures.

Peak Hour Detection

Pinpoints hours with the highest anomaly frequency, revealing operational inefficiency windows and potential night-time abnormal consumption.



Anomaly Classification

Classifies anomalies into types like Short-Term Spike, Long-Term Deviation, and Normal, aiding in distinguishing between equipment malfunction, seasonal drift, and sudden load spikes.

Business Recommendations

Based on these insights, the system generates practical recommendations such as inspecting HVAC systems during peak anomaly hours, optimizing cooling loads in summer, implementing automated monitoring alerts, adjusting operational schedules, and scheduling preventive maintenance during high-risk months.

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Deployment & Innovation Highlights

Deployment Platform: Streamlit

The system is deployed as an interactive Streamlit Dashboard, offering a user-friendly interface with several key features:

- **Model Picker:** Allows users to select their preferred anomaly detection model.
- **Cost Input:** Enables users to input the cost per energy unit for accurate financial impact calculation.
- **Real-time Training:** Facilitates on-the-fly model training and analysis.
- **Dynamic Dashboard:** Provides interactive visualizations and insights.
- **PDF Executive Report:** Generates comprehensive, automated reports.
- **Footer Branding:** Customizable branding for professional presentation.

Innovation Highlights

1

Large Dataset Handling

Efficiently processes over 800,000 records.

2

Auto Model Switching

Automatically selects optimal models for dataset size.

3

50+ Engineered Features

Comprehensive feature set for robust detection.

4

Business ROI Quantification

Translates technical findings into financial impact.

5

Executive-level PDF Reporting

Automated, detailed reports for stakeholders.

6

Interactive Dashboard

User-friendly interface for dynamic analysis.

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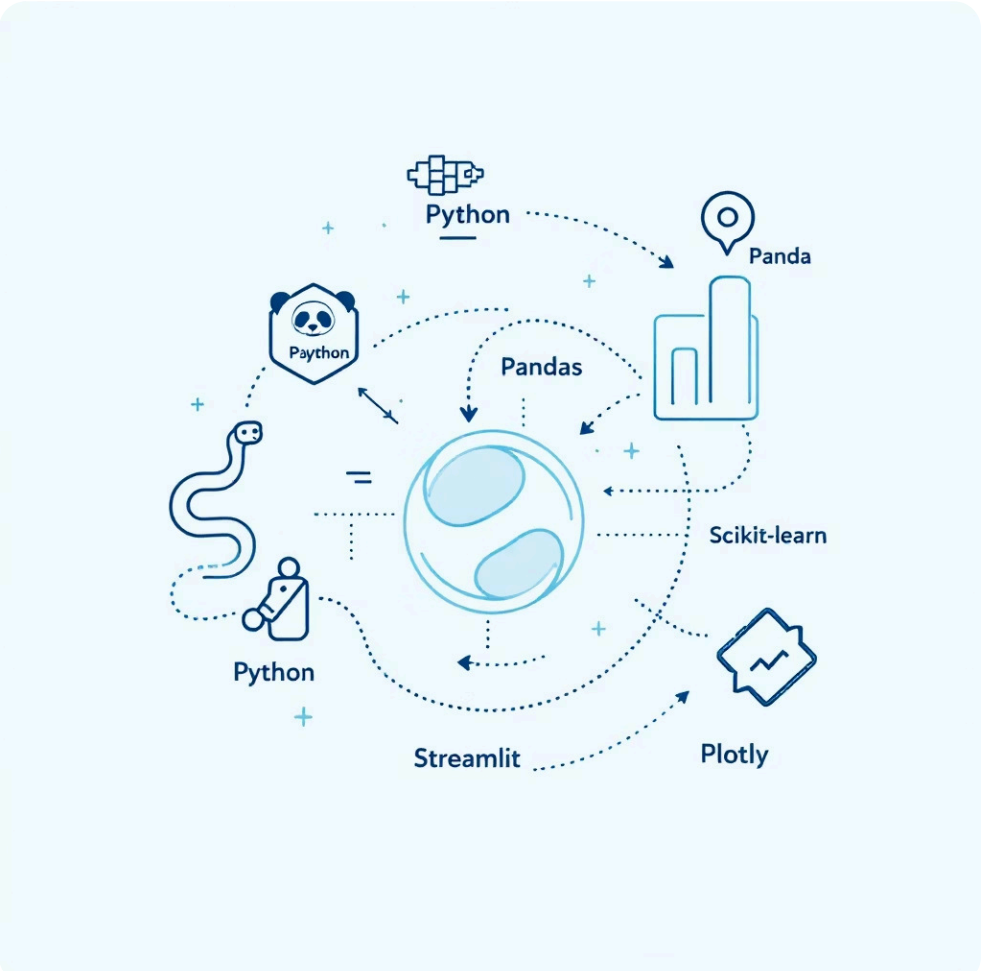
Technology Stack & Future Scope

Technology Stack

The system leverages a robust set of technologies to ensure high performance and scalability:

- **Python:** Core programming language.
- **Pandas & NumPy:** For data manipulation and numerical operations.
- **Scikit-learn:** For machine learning models.
- **Plotly & Matplotlib:** For advanced data visualization.
- **Streamlit:** For interactive dashboard deployment.
- **ReportLab:** For generating executive PDF reports.



Future Improvements

We envision several enhancements to further advance the system's capabilities:

- **Real-time Streaming Detection:** For immediate anomaly identification.
- **LSTM-based Anomaly Detection:** Incorporating deep learning for more complex patterns.
- **Auto Hyperparameter Tuning:** Optimizing model performance automatically.
- **Building-level Risk Scoring:** Providing granular risk assessments.
- **Multi-building Comparison:** Benchmarking energy performance across multiple facilities.
- **API Deployment:** Enabling seamless integration with other systems.

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Conclusion

This project successfully developed an enterprise-scale energy anomaly detection pipeline, demonstrating strong ML fundamentals, time-series expertise, and production-level deployment skills.

Comprehensive Processing

Efficiently processed over 800,000 records, achieving an approximate 5% anomaly detection rate.

Financial Quantification

Successfully quantified financial losses, providing clear ROI insights for businesses.

Actionable Insights

Delivered actionable business insights through an interactive dashboard and executive reporting.

Robust System

Showcased mastery in feature engineering and robust anomaly detection methodologies.

The system stands as a testament to our ability to transform complex data into valuable, decision-making tools, paving the way for significant energy savings and operational efficiencies in commercial buildings.

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